

18 Classification, biodiversity and conservation

A • 2026 examination • Concept guide and original exam-style practice

Understand the concept

Classification reflects evolutionary relationships using shared traits and molecular sequence data. Biodiversity can be considered at genetic, species and ecosystem levels. A high species richness alone does not describe evenness; diversity indices combine both. Conservation can protect habitats in situ and populations ex situ; maintain genetic variation to reduce vulnerability.

Worked example: follow the reasoning

Two habitats each have four species, but one has equal counts and the other is dominated by one species; the more even habitat has greater diversity by common indices.

Quick check • multiple choice

A Which pair matters for species diversity?

[1]

- A Climate and pH only
- B Richness and evenness
- C Only total individual count
- D Only DNA molecule length

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 State two reasons genetic diversity matters to a threatened population and one limitation of ex situ conservation.

[3]

Answers and explanation

Quick check: B. Richness and evenness

Question 1 [3 marks]

Different alleles improve capacity to respond to environmental change [1] and reduce inbreeding effects [1]; ex situ populations may be small/lose habitat behaviour or require reintroduction [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.